

Parametric Equations



Eliminating the parameter

- 1 Consider the curve $x = 2t + 1$, $y = t^2$.
 - a Eliminate the parameter to write y as a function of x .
 - b Identify the curve.
 - 2 Eliminate the parameter for $x = 3\cos t$, $y = 3\sin t$, $0 \leq t < 2\pi$, and describe the curve.
 - 3 For the line $x = t - 2$, $y = 2t + 3$:
 - a Find the points corresponding to $t = 0$ and $t = 2$.
 - b Eliminate the parameter to write the Cartesian equation.
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Projectile motion

- 4 A projectile follows $x = 40t$, $y = -5t^2 + 30t$ (metres, seconds).
 - a When does the projectile land ($y = 0$, $t > 0$)?
 - b Find the horizontal range.
 - c Find the maximum height.
 - 5 A second projectile is launched with $x = 25t$, $y = -5t^2 + 40t$. Find its time of flight and maximum height.
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Ellipses and other parametric curves

- 6 Eliminate the parameter for $x = 4\cos t$, $y = 2\sin t$, and identify the resulting curve.
 - 7 Eliminate the parameter for $x = t^2 - 1$, $y = t^3$ by expressing t in terms of x where possible, or by finding a relation between x and y directly.
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Finding the parametric derivative

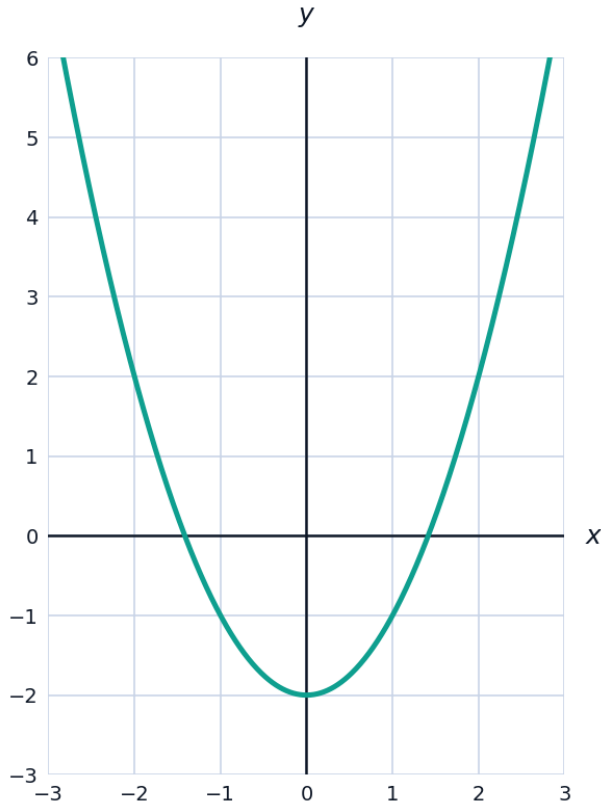
- 8 For the curve $x = t^2$, $y = 4t$, find $\frac{dy}{dx}$ in terms of t using $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$.
- 9 For the curve $x = \cos t$, $y = \sin t$, find the slope of the tangent line at $t = \frac{\pi}{4}$.

More parametric practice

- 10** A Ferris wheel of radius 10 m has its center 12 m above the ground and completes one revolution every 40 seconds. Write parametric equations for the height y of a rider above the ground as a function of time t (seconds), assuming the rider starts at the bottom of the wheel.
- 11** Two particles travel along paths $x_1 = t$, $y_1 = t^2$ and $x_2 = 4 - t$, $y_2 = 3t - 2$, sharing the same parameter t (time). Find the value of t at which they collide, and the point of collision.
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Graphing a parametric curve after elimination

- 12** For $x = t$, $y = t^2 - 2$ (with t ranging over all reals), the eliminated Cartesian form $y = x^2 - 2$ is shown. Use the graph to describe the direction the curve is traced as t increases.



Restricting the parameter domain

- 13** The curve $x = \cos t$, $y = \sin t$ for $t \in [0, \pi]$ traces only part of the unit circle. Describe which part, and state the domain and range restrictions this implies on x and y .
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Parametrizing a line segment

- 14 Write parametric equations for the line segment from $A(2, -1)$ to $B(6, 5)$, using $t \in [0, 1]$ (so $t = 0$ gives A and $t = 1$ gives B).
- 15 Using the parametrization above, find the point at $t = 0.25$ (one-quarter of the way from A to B).
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Parametric speed and motion review

- 16 A particle's position is $x(t) = 3t$, $y(t) = 4t$ for $t \geq 0$ (measured in meters, seconds). Find the particle's speed, and confirm it matches $\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$.